

Project on Machine Learning



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Question 1

1- Log-likelihood of a Gaussian

$$\log p_{\text{ID}}(x) = -\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu) - \frac{d}{2} \log(2\pi) - \frac{1}{2} \log(|\Sigma|)$$

2- Likelihood threshold and decision region

$$\begin{aligned} \log p_{\text{ID}}(x) &\geq \tau \\ \Rightarrow -\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu) - \frac{d}{2} \log(2\pi) - \frac{1}{2} \log(|\Sigma|) &\geq \tau \\ \Rightarrow (x - \mu)^T \Sigma^{-1}(x - \mu) &\leq -2\tau - d \log(2\pi) - \log(|\Sigma|) \end{aligned}$$

Let $C = -2\tau - d \log(2\pi) - \log(|\Sigma|)$. Since Σ is positive definite, the inequality describes an ellipsoid (including its interior) in feature space centered at μ .

3- Mahalanobis distance interpretation

$$(x - \mu)^T \Sigma^{-1}(x - \mu) = C$$

where $C = -2\tau - d \log(2\pi) - \log(|\Sigma|)$. This is exactly the squared Mahalanobis distance from x to μ being equal to C .

Question 2

1- Log-sum inequality

$$\log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k) \right) \neq \sum_{k=1}^K \log (\pi_k \mathcal{N}(x; \mu_k, \Sigma_k))$$

because the logarithm of a sum is not equal to the sum of the logarithms. This is due to the concavity of the logarithm.

2- Jensen's inequality and ELBO

Let $p(z = k) = \pi_k$ and $p(x|z = k) = \mathcal{N}(x; \mu_k, \Sigma_k)$. Then:

$$\log p(x) = \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x; \mu_k, \Sigma_k) \right) = \log \mathbb{E}_{p(z)}[p(x|z)]$$

By Jensen's inequality:

$$\log \mathbb{E}_{p(z)}[p(x|z)] \geq \mathbb{E}_{p(z)}[\log p(x|z)] = \sum_{k=1}^K \pi_k \log \mathcal{N}(x; \mu_k, \Sigma_k)$$

Alternatively, introducing a variational distribution $q(z)$ over the latent variable:

$$\log p(x) \geq \mathbb{E}_{q(z)} \left[\log \frac{p(x, z)}{q(z)} \right]$$

This is the Evidence Lower Bound (ELBO), which becomes tight when $q(z) = p(z|x)$. This lower bound can be written explicitly as:

$$\sum_{k=1}^K \pi_k \left[-\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) \right]$$

3- Connection to the EM algorithm

In the EM algorithm, the E-step sets the variational distribution $q(z) = p(z|x; \Theta^{(t)})$, making the bound tight. The M-step maximizes the expected complete-data log-likelihood:

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik}^{(t)} \log (\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k))$$

which is exactly the lower bound (ELBO) with respect to the parameters.

explanation:

The lower bound derived in part 2 using Jensen's inequality is exactly the ELBO:

$$\log p(x) \geq \mathbb{E}_{q(z)} \left[\log \frac{p(x, z)}{q(z)} \right] \quad (\text{ELBO})$$

In the EM algorithm, we maximize this bound in two steps:

1. **E-step:** We set $q(z) = p(z|x; \Theta^{(t)})$, the posterior distribution of the latent variable given the current parameters. This makes the bound tight (equality holds) because:

$$\log p(x) = \mathbb{E}_{p(z|x; \Theta^{(t)})} \left[\log \frac{p(x, z)}{p(z|x; \Theta^{(t)})} \right]$$

2. **M-step:** We maximize the expected complete-data log-likelihood, which is exactly the ELBO with respect to the parameters:

$$Q(\Theta, \Theta^{(t)}) = \mathbb{E}_{p(z|x; \Theta^{(t)})} [\log p(x, z; \Theta)]$$

This is equivalent to maximizing the ELBO because the entropy term

$\mathbb{E}_{p(z|x; \Theta^{(t)})} [-\log p(z|x; \Theta^{(t)})]$ does not depend on Θ .

For the GMM, the M-step objective becomes:

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik}^{(t)} \log (\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k))$$

where $\gamma_{ik}^{(t)} = p(z_i = k|x_i; \Theta^{(t)})$. This is precisely the sum of the lower bounds (over data points) derived in part 2, with $q(z)$ set to the posterior. Thus, the EM algorithm iteratively maximizes the ELBO, which is a lower bound on the log-likelihood, ensuring monotonic improvement of the true objective.

Question 3

1- Derivation of the mean update

The expected complete-data log-likelihood (Q-function) for a GMM is:

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik}^{(t)} \log(\pi_k \mathcal{N}(x_i; \mu_k, \Sigma_k))$$

where $\gamma_{ik}^{(t)} = p(z_i = k \mid x_i; \Theta^{(t)})$ are the responsibilities from the E-step. The term involving μ_k is:

$$\sum_{i=1}^n \gamma_{ik}^{(t)} \log \mathcal{N}(x_i; \mu_k, \Sigma_k) = \sum_{i=1}^n \gamma_{ik}^{(t)} \left[-\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k) \right]$$

Dropping constants with respect to μ_k , we need to maximize:

$$-\frac{1}{2} \sum_{i=1}^n \gamma_{ik}^{(t)} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k)$$

or equivalently minimize:

$$J(\mu_k) = \frac{1}{2} \sum_{i=1}^n \gamma_{ik}^{(t)} (x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k)$$

Taking the gradient with respect to μ_k and using the identity $\nabla_{\mu}(x - \mu)^T A (x - \mu) = -2A(x - \mu)$ for symmetric A :

$$\nabla_{\mu_k} J = \frac{1}{2} \sum_{i=1}^n \gamma_{ik}^{(t)} [-2 \Sigma_k^{-1} (x_i - \mu_k)] = - \sum_{i=1}^n \gamma_{ik}^{(t)} \Sigma_k^{-1} (x_i - \mu_k)$$

Setting the gradient to zero:

$$- \sum_{i=1}^n \gamma_{ik}^{(t)} \Sigma_k^{-1} (x_i - \mu_k) = 0$$

Multiplying both sides by Σ_k (which is invertible):

$$\sum_{i=1}^n \gamma_{ik}^{(t)} (x_i - \mu_k) = 0$$

Rearranging:

$$\sum_{i=1}^n \gamma_{ik}^{(t)} x_i = \mu_k \sum_{i=1}^n \gamma_{ik}^{(t)}$$

Let $N_k^{(t)} = \sum_{i=1}^n \gamma_{ik}^{(t)}$. Then:

$$\mu_k = \frac{1}{N_k^{(t)}} \sum_{i=1}^n \gamma_{ik}^{(t)} x_i$$

Thus, the update rule for the means in the M-step is obtained by maximizing the expected complete-data log-likelihood with respect to μ_k .

2- Gradient-based optimization

$$\nabla_{\mu_k} Q = \sum_{i=1}^n \gamma_{ik}^{(t)} \Sigma_k^{-1} (x_i - \mu_k)$$

Setting to zero:

$$\begin{aligned} \sum_{i=1}^n \gamma_{ik}^{(t)} (x_i - \mu_k) &= 0 \\ \Rightarrow \mu_k \sum_{i=1}^n \gamma_{ik}^{(t)} &= \sum_{i=1}^n \gamma_{ik}^{(t)} x_i \\ \Rightarrow \mu_k^{(t+1)} &= \frac{1}{N_k^{(t)}} \sum_{i=1}^n \gamma_{ik}^{(t)} x_i, \quad N_k^{(t)} = \sum_{i=1}^n \gamma_{ik}^{(t)} \end{aligned}$$

3- Interpretation of the update

The update is a weighted average of all data points x_i , where each weight $\gamma_{ik}^{(t)}$ represents the probability that x_i belongs to component k . This pulls the mean toward points that are likely to have been generated by that component.

Question 4

1- Expected log-likelihood in high dimensions

For isotropic Gaussian $\mathcal{N}(\mu, \sigma^2 I)$:

$$\log p(x) = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|x - \mu\|^2$$

If $x \sim \mathcal{N}(\mu, \sigma^2 I)$, then $\|x - \mu\|^2 / \sigma^2 \sim \chi^2(d)$, so $\mathbb{E}[\|x - \mu\|^2] = d\sigma^2$. Thus:

$$\mathbb{E}[\|x - \mu\|^2] = \mathbb{E}\left[\sum_{j=1}^d (x_j - \mu_j)^2\right] = \sum_{j=1}^d \mathbb{E}[(x_j - \mu_j)^2] = d\sigma^2$$

$$\mathbb{E}[\log p(x)] = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \cdot d\sigma^2 = -\frac{d}{2} \log(2\pi\sigma^2) - \frac{d}{2}$$

2- OOD samples with high likelihood

The log-likelihood depends on the squared distance $\|x - \mu\|^2$. An OOD sample x_{OOD} may lie closer to μ than an ID sample x_{ID} does, especially if:

- The ID sample is atypically far from the mean (possible in high dimensions due to the Gaussian annulus phenomenon).
- The OOD sample is structured to have a small distance to μ despite being from a different distribution.

Thus, even though x_{OOD} is OOD, it may have a higher likelihood under the ID model if its Mahalanobis distance to μ is smaller.

3- Failure of likelihood-based OOD detection

In high dimensions, the Gaussian density concentrates in a thin shell of radius $\sim \sqrt{d}\sigma$. Both ID and OOD samples can have similar distances to the mean, leading to overlapping likelihood distributions. Moreover, some OOD samples may fall inside high-density regions of the ID model (e.g., near the mean), yielding high likelihoods. Conversely, some ID samples may fall in low-density tails. This overlap makes likelihood thresholding unreliable for OOD detection in high-dimensional spaces. Additional mechanisms (e.g., likelihood ratio tests, density reweighting, or feature-space methods) are needed to improve robustness.